Problem sheet–7

1. Using duality between Continuous-time Fourier Series and Discrete-time Fourier Transform, prove

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi}\int_{2\pi} \left|X(e^{j\Omega})\right|^2 d\Omega$$

2. If the inverse F.T. of $Y(e^{jw})$ is given by

$$y[n] = \left(\frac{\sin(w_c n)}{\pi n}\right)^2$$

where $0 < w_c < \pi$. Determine the value of w_c that ensures $Y(e^{jw}) = 0.25$.

3. Given the fact that DTFT$\{0.8^n u[n] = 1/(1-0.8\, e^{-jw})\}$ and using the properties, compute the DTFT of the following sequences:

a) $x[n] = 0.8^n u[n-2]$

b) $x[n] = 0.8^n u[n] \cos(0.1\pi n)$

c) $x[n] = n\, 0.8^n u[n]$

d) $x[n] = 0.8^n u[-n]$

e) $x[n] = \begin{cases} 0.8^n & \text{if } 0 \le n \le 5 \\ 0 & \text{otherwise} \end{cases}$

4. Using the definition determine the DTFT of the following sequences. It it does not exist say why:

a) $x[n] = 0.5^n u[n]$

b) $x[n] = 0.5^{|n|}$

c) $x[n] = 2^n u[-n]$

d) $x[n] = 0.5^n u[-n]$

e) $x[n] = 2^{|n|}$

f) $x[n] = 3(0.8)^{|n|} \cos(0.1\pi n)$

Here u[n] is a unit step function.

5. Difference equations

Consider an LTI system that is characterized by the difference equation:

$$y[n] = \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = 2x[n]$$

a) Find the impulse response of the above system.
b) Why did we not require any auxiliary conditions for solving the impulse response?
c) Does system stability influence our solutions?

6. A particluar discrete-time system has input x[n] and output y[n]. The Fourier transforms of these signals are related by the following equation:

$$Y(\Omega) = 2X(\Omega) + e^{-j\Omega}X(\Omega) - \frac{dX(\Omega)}{d\Omega}$$

a) Is the system linear? Clearly justify your answer.
b) Is the system time-invariant?Clearly justify your answer.
c) what is y[n] if x[n] = $\delta[n]$?

7. Let $\tilde{x}[n]$ be a periodic signal with period N. A finite-duration signal x[n] is related to $\tilde{x}[n]$ through

$$x[n] = \begin{cases} \tilde{x}[n], & n_0 \le n \le n_0 + N - 1 \\ 0, & otherwise \end{cases}$$

for some integer n_0. That is x[n] is equal to $\tilde{x}[n]$ over one period and zero elsewhere.

a. If $\tilde{x}[n]$ has Fourier series coefficients a_k and x[n] has Fourier transform $X(e^{jw})$, show that

$$a_k = \frac{1}{N} X(e^{j2\pi k/N})$$

Regardless of the value of n_0

b. Consider the following two signals:

$$x[n] = u[n]-u[n-5]$$

$$\tilde{x}[n] = \sum_{k=-\infty}^{\infty} x[n-kN]$$

Where N is a positive integer. Let a_k denote the Fourier series $\tilde{x}[n]$ and let $X(e^{jw})$ coefficients of denote the Fourier transform of x[n].

I. Determine a closed-form expression for $X(e^{jw})$

II. Using the result of part (i), determine an expression for the Fourier coefficients a_k